

According to the program code in Fig. 1, if we made an instruction dependency graph
?What is the number of steps that can be executed in parallel

* 1 point

PC	Instruction
00	a = 2
01	b = 4
02	tmp2 = a + b // 6
03	tmp3 = tmp2 + a // 8
04	tmp4 = b + b // 8
05	tmp5 = b * b // 16
06	tmp6 = tmp2 + tmp4 // 14
07	tmp7 = tmp5 + tmp6 // 30
08	if (tmp3 > 7)
09	print tmp3
	else
10	print tmp7

☐ 4

☐ 3

☐ 6

☐ 2

According to the program code in Fig. 1, if we made an instruction dependency graph

* 1 point

?What is the minimum number of clock cycles required

PC	Instruction
00	a = 2
01	b = 4
02	tmp2 = a + b // 6
03	tmp3 = tmp2 + a // 8
04	tmp4 = b + b // 8
05	tmp5 = b * b // 16
06	tmp6 = tmp2 + tmp4 // 14
07	tmp7 = tmp5 + tmp6 // 30
08	if (tmp3 > 7)
09	print tmp3
	else
10	print tmp7

☐ 8

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☐ 9

type of the cache is located directly within the processor ...

L1 ☒

L2 ☐

L3 ☐

L2 & L3 ☐

السؤال التالي <

> السؤال السابق

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A multi-core chip with 16 cores and 8 arithmetic ALUs in each core that operate in a SIMD fashion with 6 concurrent threads inside each core, what is the number of parallel tasks required to fill the chip's capacity and ensure no ALU sits idle during a stall

512

768

1024

128

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Accessing memory is a major source of stalls

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صح



خطأ



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Assume in the running program, the threads perform three arithmetic instructions, followed by memory load with 12 cycle latency, what is the best number of threads for 100% processor utilization

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If there is no dependency in the program, the Superscalar processor performance is similar to simple processor performance

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صح

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السؤال التالي <

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:In SIMD computing, which is correct

fetches and decodes one instruction ☐

process multiple pieces of data at the same time ☐

allow many Execution Units (ALUs) simultaneously ☐

All of them ☒

السؤال التالي <

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9. Superscalar processor can decode and execute up to two instructions per clock

صح ☒

خطأ ☐

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Superscalar processor executes tasks in a single instruction stream

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صح



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The instruction execution is mainly changing the computer's state

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صح



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السؤال التالي <

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The usage of cache can change the output of a program

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صح ☐

خطأ ☒

السؤال التالي >

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What is cache action when the requested data is exist in the cache

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Hit



Cold miss



Loading



None of them



السؤال التالي <

> السؤال السابق

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What is the memory latency If the amount of time it takes the memory system to provide
?data to the processor is 0.00005 s, and provide 4 items in the same time

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≤ 0.00005



≤ 0.0002



≤ 0.5



≤ 005.0



> السؤال السابق

السؤال التالي <

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Parallel Computing Midterm

الحوسبة الموزعة

46:10 ⌚

الوقت المتبقي

السؤال 15 من 15

تم الإجابة على 14 سؤال

When using ISPC compiler work on programmer sequential code, use SIMD units to vectorizes it, allowing speedup

15

صح ☒

خطأ ☐

تسليم الاختبار

السؤال السابق >

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